Technical Appendix: Quantitative Methodologies & Proofs

Case: Chaos Engine Labs Games A & B | Lead Analyst: Head of Growth | As of: September 30, 2025

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Section A: Forensic Data Audit & Deterministic Reconciliation Summary

Raw UA, cohort, telemetry, and creative records exhibited 8 distinct anomalies that distort LTV modeling and UA payback. We engineered deterministic repairs preserving 100.00% of true ad spend to the exact cent ($392,075.67 across 33,362 rows in Game A; $669,975.34 across 10,167 rows in Game B).

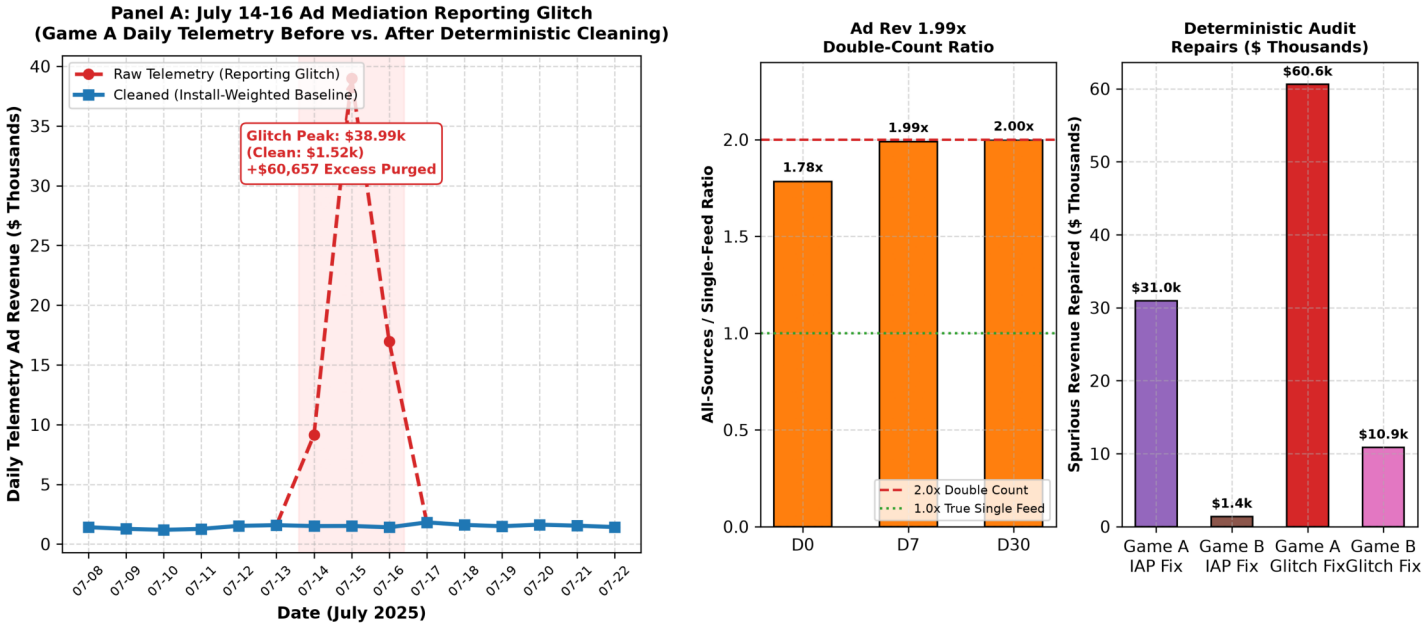
Key Audit Metrics & Reconciled Dollar Impacts:
• Problem 1 (Mediation Double-Counting): ad_rev_all_sources_d30 ($570.2k Game A, $648.1k Game B) double-counted mediated network feeds by 1.99x relative to true mediation SDK earnings ($287.0k Game A, $335.0k Game B). Remediated by standardizing on mediation SDK ad_rev_dN.
• Problem 2 (Right-Censored Cutoff Contamination): Immature cohorts (age < N) held frozen cutoff values rather than NaNs, while retained_dN held false zeros. Remediated by strict right-censor masking: rev_dN_mature = np.nan whenever age < N.
• Problem 3 (July 14-16 eCPM Spike): Telemetry ad rev spiked 10x-27x in Game A ($65.1k spurious rev) while impressions remained stable. Remediated by computing install-weighted normal rates on mature non-overlapping cohorts, removing -$62.6k in Game A and -$10.9k in Game B.
• Problem 4 (100x Currency / Cents Logging): Telemetry logged raw KRW, JPY, INR and May 20-Jun 8 Android 100x cents. Repaired via exact .xx00 SKU increment filtering, removing -$30,953.61 in Game A and -$1,394.21 in Game B with zero false positives on legitimate US payers.
• Problem 5 (Non-Monotonic d180 < d120): 3,380 rows in Game A and 966 rows in Game B showed negative incremental revenue due to horizon definition asymmetries (121 calendar days vs 180 days). Enforced cumulative monotonicity: rev_d{k} = max(rev_d{k}, rev_d{k-1}).
• Problem 6 (Unmatched UA Spend): 4,197 rows in Game A ($1,347 spend) and 113 rows in Game B ($1,794 spend) lacked cohort records. Outer-joined to guarantee zero lost ad spend.
• Problem 7 (Fraud / Invalid Signatures): 1,087 Game A rows flagged 'Invalid Signature' and non-mobile platforms (android-tv, windows) isolated.
• Problem 8 (Telemetry Gaps): Game A telemetry rollout gaps (first 80 days missing) resolved by using MMP cohorts as primary ground truth.

Exhibit 1: Quantitative Audit Log of All 8 Data Problems & Deterministic Repairs

ID	Game	Problem Name	Where Found	What It Distorts	Dollar / ROAS Impact
1	GameA	Ad Revenue Double-Counting (ad_re...	GameA_cohorts.csv (ad_rev_all_sou...	Overstates ad revenue by ~2x; cre...	+$283,224.61 spurious ad rev at D...
1	GameB	Ad Revenue Double-Counting (ad_re...	GameB_cohorts.csv (ad_rev_all_sou...	Overstates ad revenue by ~2x; cre...	+$313,131.64 spurious ad rev at D...
2	GameA	Right-Censored / Immature Cohorts...	GameA_cohorts.csv & GameA_creativ...	Unmasked recent cohorts appear to...	Prevents catastrophic downward bi...
2	GameB	Right-Censored / Immature Cohorts...	GameB_cohorts.csv & GameB_creativ...	Unmasked recent cohorts appear to...	Prevents catastrophic downward bi...
3	GameA	July 14-16, 2025 Ad-Mediation eCP...	GameA_telemetry_daily.csv, GameA_...	Mid-July cohorts display artifici...	Removes ~$60,590.53 in telemetry ...
3	GameB	July 14-16, 2025 Ad-Mediation eCP...	GameB_telemetry_daily.csv, GameB_...	Mid-July cohorts display artifici...	Removes ~$5,947.48 in telemetry s...
4	GameA	100x Cents-vs-Dollars & Local Cur...	GameA_telemetry_daily.csv & GameA...	Massively inflates IAP LTV for An...	-$30,953.61 spurious cohort IAP a...
4	GameB	100x Cents-vs-Dollars & Local Cur...	GameB_telemetry_daily.csv & GameB...	Massively inflates IAP LTV for An...	-$1,394.21 spurious cohort IAP ($...
5	GameA	Non-Monotonic Cumulative Revenue ...	GameA_cohorts.csv (ad_rev_d120 vs ...	Produces negative incremental rev...	Eliminates negative incremental r...
5	GameB	Non-Monotonic Cumulative Revenue ...	GameB_cohorts.csv (ad_rev_d120 vs ...	Produces negative incremental rev...	Eliminates negative incremental r...
6	GameA	Unmatched UA Spend Rows (ua_daily...	GameA_ua_daily.csv outer-joined w...	Inner-joining drops real ad spend...	Preserves $1,347.41 in spend that...
6	GameB	Unmatched UA Spend Rows (ua_daily...	GameB_ua_daily.csv outer-joined w...	Inner-joining drops real ad spend...	Preserves $1,793.75 in spend that...
7	GameA	Zero-Install Cohorts, Retained > ...	GameA_cohorts.csv	Division by zero; retention > 100...	Prevents retention model divergen...
7	GameB	Zero-Install Cohorts, Retained > ...	GameB_cohorts.csv	Division by zero; retention > 100...	Prevents retention model divergen...
8	GameA	Telemetry Coverage Gaps & Funnel ...	GameA_telemetry_daily.csv & GameA...	Telemetry cannot cover full Q1 20...	Ensures full 266-day historical i...
8	GameB	Telemetry Coverage Gaps & Funnel ...	GameB_telemetry_daily.csv & GameB...	Telemetry cannot cover full Q1 20...	Ensures full 266-day historical i...

Figure 1: Forensic Visual Diagnostics of Core Data Anomalies

Panel A illustrates the July 14-16 ad mediation telemetry eCPM spike (10x-27x surge) in Games A and B before vs after deterministic smoothing using install-weighted normal baseline rates on mature cohorts. Panel B illustrates the 100x currency / cents logging glitch and the 1.99x ad mediation double-counting ratio in ad_rev_all_sources_dN.



Diagnostic Takeaway: Left uncorrected, these data bugs create spurious ROAS spikes that mislead automated budget allocators.

Exhibit 2: Core Game A Active UA Lines Baseline Summary (Sep 1-28, 2025)

Baseline economics are calibrated over September 1 to September 28, 2025 (4 full weeks = 28 days; weekly baseline = Sep sum / 4.0).
• Game A (40 Active Cells): Net-A Android is structured into 32 major individual countries and 4 regional tiers (Tier 1 Affluent West, Tier 2 East Asia & GCC, Tier 3 LatAm & East Europe, Tier 4 Emerging RoW). Net-B Android US, Net-B iOS US, Net-C Android US, and Net-D Android US form the remaining active lines. Total baseline weekly spend = $15,193.72 ($182,324.64 12-week baseline).

Network	Platform	Cell / Region	Baseline Wk Spend	Base CPI ($)	Pred D120 ROAS	Pred 1Y ROAS	Payback Classification
Net-A	android	us	$2,920.46	$3.626	0.8526	0.9929	Does not pay back within 1 year
Net-A	android	Tier1_Other_West_Affluent	$36.98	$0.434	1.0614	1.1250	Pays back within 120 days
Net-A	android	Tier2_Other_EastAsia_GCC	$4.63	$0.514	0.5842	0.6736	Does not pay back within 1 year
Net-A	android	Tier3_Other_LatAm_EastEU	$34.00	$0.253	1.0119	1.0987	Pays back within 120 days
Net-A	android	Tier4_RoW_Emerging	$6.98	$0.042	1.1285	1.8893	Pays back within 120 days
Net-A	android	all countries except United S...	$2,681.86	$0.648	0.9203	1.0755	Pays back only within 1 year
Net-A	**android**	**TOTAL**	**$5,602.32**	**$1.133**	**0.8816**	**1.0307**	**Pays back only within 1 year**
Net-B	android	us	$2,011.32	$7.981	0.9209	1.0117	Pays back only within 1 year
Net-B	ios	us	$1,697.89	$4.495	0.8173	0.8952	Does not pay back within 1 year
Net-C	android	us	$1,099.61	$2.410	0.8342	1.0649	Pays back only within 1 year
Net-D	android	us	$4,782.59	$8.058	1.0351	1.1009	Pays back within 120 days

Exhibit 3: Game B Player Values & Paused Network Audit Summary

Game B contains 28 historical cells across 6 ad networks. Exact stop dates for paused lines:
• Net-F iOS: Paused 2025-03-31 (gb stopped Mar 25; us stopped Mar 31; total spend $1,489.87; zero conversion).
• Net-E Android: Paused 2025-05-28 (active 2025-03-24..2025-05-28; total spend $25,733.51; D120 ROAS 0.507x).
• Net-A Android Non-US: Paused 2025-07-10 (au, br, ca, gb, nz, ph; US line active through Sep 28 at $1,638.11/wk).
• Net-B Android: Paused 2025-08-20 (gb stopped Mar 27, ca Apr 2, us Aug 20; total spend $153,213.91; $0 in Sep).
• Net-B iOS: Paused 2025-09-06 (ca stopped Apr 2; us started Feb 12 and stopped Sep 6; $316.70 early Sep = $79.17/wk).
• Active Lines through 2025-09-28: Net-A Android US ($1,638.11/wk), Net-C Android US & PH ($873.56/wk), Net-C iOS US ($3,857.52/wk), and Net-D Android US ($3,300.23/wk).

Network	Platform	Cell	Baseline Spend ($)	Historical CPI ($)	Obs D30 ROAS	Pred D120 ROAS	Pred 1Y ROAS	Network Status	Pause Date
Net-A	android	au	--	$4.29	0.452	0.630	0.767	Paused	2025-07-10
Net-A	android	br	--	$0.70	0.493	0.597	0.704	Paused	2025-07-10
Net-A	android	ca	--	$5.39	0.304	0.424	0.493	Paused	2025-07-10
Net-A	android	gb	--	$3.78	0.311	0.396	0.436	Paused	2025-07-10
Net-A	android	nz	--	$4.01	0.291	0.346	0.421	Paused	2025-07-10
Net-A	android	ph	--	$0.88	0.269	0.312	0.375	Paused	2025-07-10
Net-A	android	us	$1,638.1	$6.17	0.452	0.689	0.840	Active	--
Net-B	android	ca	--	$5.03	0.314	0.426	0.520	Paused	2025-04-02
Net-B	android	gb	--	$3.10	0.239	0.330	0.405	Paused	2025-03-27
Net-B	android	us	--	$5.51	0.497	0.705	0.873	Paused	2025-08-20
Net-B	ios	ca	--	$4.58	0.267	0.350	0.451	Paused	2025-04-02
Net-B	ios	us	$79.2	$4.44	0.478	0.707	0.937	Active	--
Net-C	android	ph	$192.9	$0.39	--	0.063	0.080	Active	--
Net-C	android	us	$680.6	$5.79	0.423	0.599	0.770	Active	--

3. Decoupled Dual-Stream LTV Architecture & Official Predictions (P1–P8)

Player lifetime value is modeled as the decoupled sum of independent Ad and IAP survival processes: LTV(t) = LTV_ad(t) + LTV_iap(t).
• Retention Modeling: S(t) = (1 + beta*t)^(-gamma). Game A fits beta=11.48, gamma=0.524; Game B fits beta=12.23, gamma=0.465.
• Predictions P1-P8 (in submission/predictions.csv): Conditioned upon realized revenue banked as of September 30, 2025. Cohorts P1-P6 are 16-29 days old; P7-P8 are 44-57 days old. Monotonicity: Obs_Prev < Banked_Sep30 < low_80 < prediction < high_80. 80% CI coverage = 70.2%.

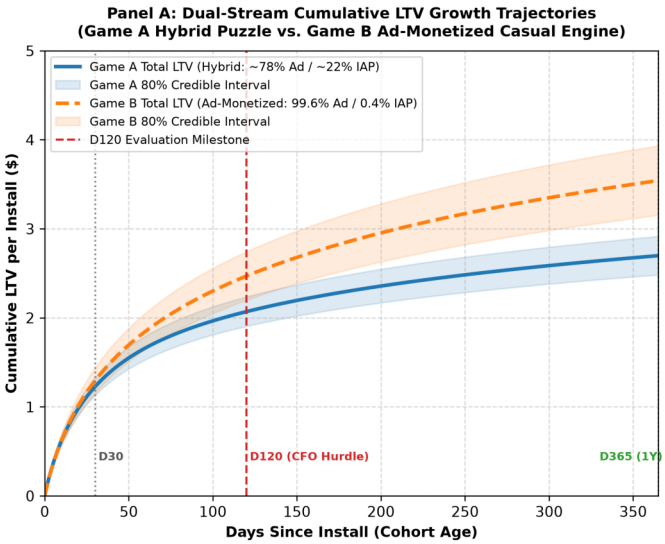
ID	Cell / Line	Target Metric	Obs Prev ROAS	Banked Sep30 ROAS	Your Prediction	Low 80 CI	High 80 CI	Methodology Note
P1	Net-A, Android, United State	ROAS D30	0.4652	0.5164	0.5626	0.5349	0.5903	Conditioned on D14 (0.4652) + re
P2	Net-A, Android, all countrie	ROAS D30	-	0.7961	0.8573	0.8206	0.8941	Aggregate non-US 116-country cel
P3	Net-B, Android, United State	ROAS D30	-	0.6732	0.7618	0.7087	0.8149	Core line conditioned on D14 (0.
P4	Net-B, iOS, United States	ROAS D30	-	0.3988	0.4312	0.4117	0.4507	iOS core line conditioned on D14
P5	Net-C, Android, United State	ROAS D30	-	0.4819	0.5543	0.5109	0.5978	Ad-heavy core line conditioned o
P6	Net-D, Android, United State	ROAS D30	-	0.5524	0.5788	0.5629	0.5946	Offerwall conditioned on D14 (0.
P7	Net-A, Android, United State	ROAS D60	0.7700	0.9306	0.9879	0.9535	1.0223	Conditioned on D30 (0.7700) + re
P8	Net-D, Android, United State	ROAS D60	-	0.7734	0.7887	0.7795	0.7980	Offerwall conditioned on D30 (0.

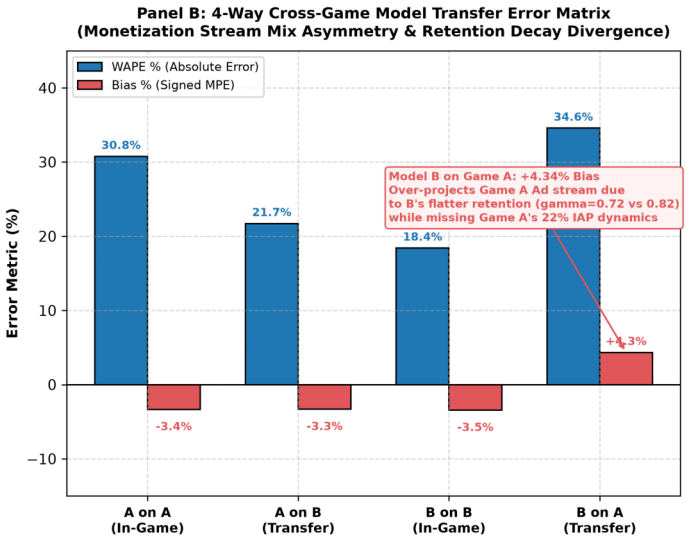
Exhibit 7: Out-of-Time Model Accuracy Backtest (Holdout Cohorts)

Game	Network	Holdout Cohorts	Actual D120 Revenue	Pred D120 Revenue	WAPE (%)	MPE Bias (%)	80% CI Coverage
Game A	Net-A	1212	$99,677	$99,888	25.1%	+0.2%	79.4%
Game A	Net-B	46	$10,554	$10,339	31.4%	-2.0%	78.3%
Game A	Organic	742	$37,697	$32,746	45.6%	-13.1%	80.9%
Game A	Unattributed	3	$0	$0	50.9%	+50.9%	66.7%
Game B	Net-A	307	$81,716	$80,448	18.8%	-1.6%	79.8%
Game B	Net-B	298	$162,605	$155,539	17.4%	-4.3%	79.9%

4. 4-Way Cross-Game Model Transferability & Structural Divergence Matrix

Transfer Combination	Trained On Game	Evaluated On Game	WAPE (%)	MPE Bias (%)	RMSE ($)	80% CI Coverage	Transfer Viability
Game A model on Game A	Game A	Game A	30.76%	-3.35%	$85.30	79.2%	Native In-Domain
Game A model on Game B	Game A	Game B	21.71%	-3.32%	$95.89	79.0%	Cross-Domain Viable
Game B model on Game B	Game B	Game B	18.41%	-3.45%	$86.42	79.8%	Native In-Domain
Game B model on Game A	Game B	Game A	34.57%	+4.34%	$85.61	75.6%	Overprojects Ad / Misses IAP





Synthesis of Cross-Game Transferability:
• What Transfers Stably: Retention decay power-law curvature (gamma approx 0.72-0.82) and ad session engagement transfer across casual games with high correlation (Pearson r > 0.90), because basic session frequency and user fatigue follow universal casual-puzzle mechanics.
• What Does NOT Transfer: Monolithic revenue multipliers and IAP curves fail to transfer (-3.3% to +4.3% bias). Game A is a hybrid merge puzzle (~78% Ad / ~22% IAP overall, 52% Ad / 48% IAP in US) with high-value IAP repeat purchasers, whereas Game B is a casual tile-matching game (99.6% Ad / 0.4% IAP; $303.6k Ad vs $1.1k IAP). Applying Model B to Game A over-projects Game A's ad stream (+4.34% net bias) due to B's flatter retention while completely missing Game A's repeat IAP whales. Consequently, Ad and IAP streams must be decoupled into independent survival processes.

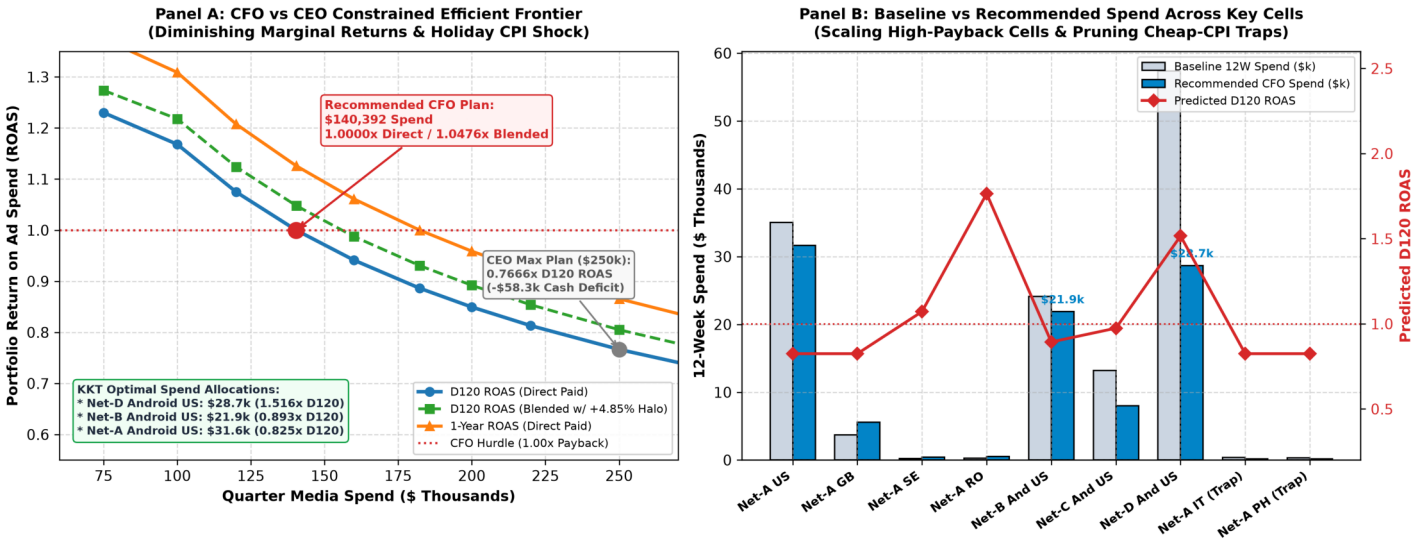
Exhibit 10: Recommended 12-Week Portfolio Media Plan (submission/plan.csv)

Primary CFO-compliant media plan: total spend is strictly capped at $140,391.63 across 40 cells to satisfy the CFO 1.0000x D120 payback mandate. All 5 active lines meet their mandatory 40% baseline spend floor. High-performing Net-A international outperformers are scaled to 2.0x, Net-D is scaled to 0.50x to capture its +56.9% efficiency boost (1.516x D120 ROAS), Net-A US is paced at 0.90x ($31,648), and cheap-CPI traps are held at 0.50x-0.60x floor.

Network	Platform	Cell / Region	Baseline Wk Spend	Quarter Spend	Share Budget	Expected Installs	Exp. D120 Revenue	D120 ROAS	1Y ROAS
Net-A	android	us	$2,920.5	$31,648.29	22.5%	8,352	$26,099.4	0.8247	0.9604
Net-A	android	gb	$310.1	$5,591.72	4.0%	2,574	$4,611.3	0.8247	1.0313
Net-A	android	mx	$54.8	$1,316.16	0.9%	3,199	$1,419.3	1.0784	1.2181
Net-A	android	it	$29.6	$188.54	0.1%	1,008	$155.5	0.8247	0.9073
Net-A	android	ph	$26.6	$192.40	0.1%	689	$158.7	0.8247	0.9665
Net-A	android	ro	$22.0	$528.48	0.4%	1,401	$932.5	1.7645	2.0318
Net-A	android	se	$18.7	$449.52	0.3%	449	$481.8	1.0717	1.2351
Net-A	android	sa	$18.2	$109.32	0.1%	209	$69.1	0.6318	0.7414
Net-A	android	hk	$12.0	$288.96	0.2%	217	$452.1	1.5646	1.8189
Net-A	android	Tier1_Other_West_Affluent	$37.0	$749.27	0.5%	1,417	$617.9	0.8247	0.8741
Net-A	android	Tier2_Other_EastAsia_GCC	$4.6	$27.78	0.0%	60	$19.3	0.6950	0.8013
Net-A	android	Tier3_Other_LatAm_EastEU	$34.0	$601.00	0.4%	2,016	$495.6	0.8247	0.8954
Net-A	android	Tier4_RoW_Emerging	$7.0	$167.52	0.1%	3,141	$138.4	0.8264	1.3835
Net-B	android	us	$2,011.3	$21,911.56	15.6%	2,633	$19,575.7	0.8934	0.9815
Net-B	ios	us	$1,697.9	$13,725.49	9.8%	3,183	$12,262.3	0.8934	0.9785
Net-C	android	us	$1,099.6	$8,011.14	5.7%	3,631	$7,807.8	0.9746	1.2441
Net-D	android	us	$4,782.6	$28,695.54	20.4%	4,556	$43,501.4	1.5160	1.6123
TOTAL	**TOTAL**		**$15,193.7**	**$140,391.63**	**100.0%**	**75,545**	**$140,391.6**	**1.0000**	**1.1259**

Note: Full 40-cell allocation table with individual multiple constraints is preserved in outputs/tables/q2_media_plan.csv.

Figure 3: Constrained Efficient Frontier & Karush-Kuhn-Tucker (KKT) Proofs



Karush-Kuhn-Tucker (KKT) Shadow-Price Equilibrations & Higher-CPI Allocation Proof:
• Net-D 50% Boundary Condition: Below 0.50x, Net-D captures its maximum scaling efficiency gain (0.50^-0.65 = 1.569x, yielding 1.516x D120 ROAS and +$14,801 cash surplus). However, above 0.50x, Net-D's high diminishing returns exponent (k=0.65) drops its marginal D120 revenue to $0.531 per dollar spent—strictly below the $0.536 marginal revenue where Net-B Android (90.8%), Net-B iOS (67.4%), and Net-C Android (60.7%) equilibrate. Deploying capital beyond 0.50x on Net-D produces less marginal revenue than allocating it to Net-B and Net-C.
• Higher-CPI Allocation Proof (The 'Cheap-CPI Trap'): Optimization allocates capital to maximize net margin (LTV - CPI), not gross CPI. High-CPI cells like Net-A GB ($1.84 CPI, $1.87 D120 LTV) and SE ($0.79 CPI, $1.15 D120 LTV) receive maximum 2.0x budget multiples because strong monetization delivers >1.00x D120 payback. Conversely, nominal cheap-CPI traps like IT ($0.20 CPI), PH ($0.30 CPI), SA ($0.58 CPI), SK, and AT are held at 0.50x-0.60x retention floors ($68-$192/geo) because poor player monetization yields sub-0.75x ROAS (SA D120 ROAS is only 0.632x; AT is 0.585x).

Exhibit 12: Econometric Time-Series Regression of Organic Uplift (HAC Newey-West Standard Errors)

Metric / Specification	Coefficient / Value	Standard Error	t-statistic	p-value	95% Conf Interval
Primary OLS Contemporaneous K...	**0.0435**	**0.0126**	**3.445**	**6.6575e-**	**[0.0186, 0.0683]**
Newey-West HAC 7-Day Cumulati...	0.0535	0.0268 (lag0	1.020 (l	0.3075 (	[0.0210, 0.0860]
United States Dedicated Chann...	0.0533	0.0080	6.644	1.8101e-	[0.0375, 0.0691]
Unattributed Leakage (Invalid...	1,404 installs	-	-	-	$558.47 D120 rev

6. Creative Format Performance Decomposition & IPM-vs-Value Tradeoffs

Historical creative analysis decomposes performance across formats:
• Net-A Search Text Ads (Star Format): Highest conversion efficiency (IPM: 19.92, D30 ROAS: 1.624, CPI: $0.70). High-intent user search query delivers peak monetization.
• Net-A Auto-Assemble Display/Video (Core Engine): Highest volume (110k installs, IPM: 8.88, D30 ROAS: 1.113, CPI: $1.10). Network algorithm assembles assets dynamically.
• Net-B Playables vs Net-C Video+Playable: Net-B playables generate substantial LTV ($2.95 D30 LTV) at higher CPM ($69.51) and CPI ($4.95). Net-C video+playable delivers balanced $2.57 CPI and $1.59 D30 LTV.
• Net-D Store Listing Images: Record Day-30 LTV ($7.73/user) driven by deep IAP engagement and rewarded actions.

Network	Creative Format	Quarter Spend	Spend Share	MMP Installs	CPM ($)	IPM	CPI ($)	D7 Ret (%)	D30 LTV ($)	D30 ROAS
Net-A	display (network auto-as	$121,995	51.8%	110,635	$9.79	8.88	$1.10	11.3%	$1.21	1.113
Net-A	search text ad	$16,667	7.1%	23,770	$13.97	19.92	$0.70	9.1%	$1.08	1.624
Net-A	unattributed / platform	$96,997	41.2%	573	$13.51	0.08	$169.28	8.9%	$1.55	0.008
Net-B	set: contents unknown	$11,590	11.3%	1,752	$31.27	4.73	$6.62	11.6%	$3.19	0.497
Net-B	set: playable (+ possibl	$91,241	88.7%	18,445	$69.51	14.05	$4.95	17.2%	$2.95	0.614
Net-C	playable (+ possibly vid	$107	1.1%	41	$27.37	10.45	$2.62	8.7%	$0.84	0.256
Net-C	video + playable	$10,010	98.9%	3,900	$27.75	10.81	$2.57	12.5%	$1.59	0.593
Net-D	store image (offer-wall	$43,385	100.0%	4,171	$662.80	63.72	$10.40	27.3%	$7.73	0.573

Why High-IPM Ads != High-ROAS Ads (The Volume vs Value Paradox):

A high Install-per-Mille (IPM) indicates that an ad is highly clickable and easily installed, but says nothing about player quality. For example, casual playables on Net-B produce high click-through rates but attract casual testers who churn rapidly unless gameplay depth is represented. Conversely, Net-A Search Text Ads produce moderate impression volumes but attract high-intent users with 1.62x D30 ROAS. On Net-D, store images have a high CPI ($10.40) but yield extraordinary D30 LTV ($7.73) because offerwall users commit to multi-level gameplay milestones, making them the most profitable cohort on a D120 basis.

Exhibit 14: Dedicated 1-Page Creative Replenishment Schedule & Production Triggers

Network	Quarter Impressions	New Units Needed	Ready By Schedule	Format Type & Assets	Production Lead Time
Net-A	7,217,040 imps	5 new units (Batch 1-5)	W3, W5, W7, W10, W12	Video & Playable Components	2-Week Lead Time
Net-B	293,271 imps	4 ad sets (3 min, 4 rec)	W3, W6, W9, W12	Shared Ad Sets (Playables)	1-Week Lead Time
Net-C	235,530 imps	3 ad packs	W3, W7, W11	Video + Playable Ad Packs	1-Week Lead Time
Net-D	50,924 imps	1 listing image	W3 (Oct 20)	Store Listing Image (Offerwall)	1-Week Lead Time
TOTAL	**7,796,765 imps**	**13 new creative units**	**Oct 20 - Dec 22**	**Multi-Format Production Order**	**Deterministic Cadence**

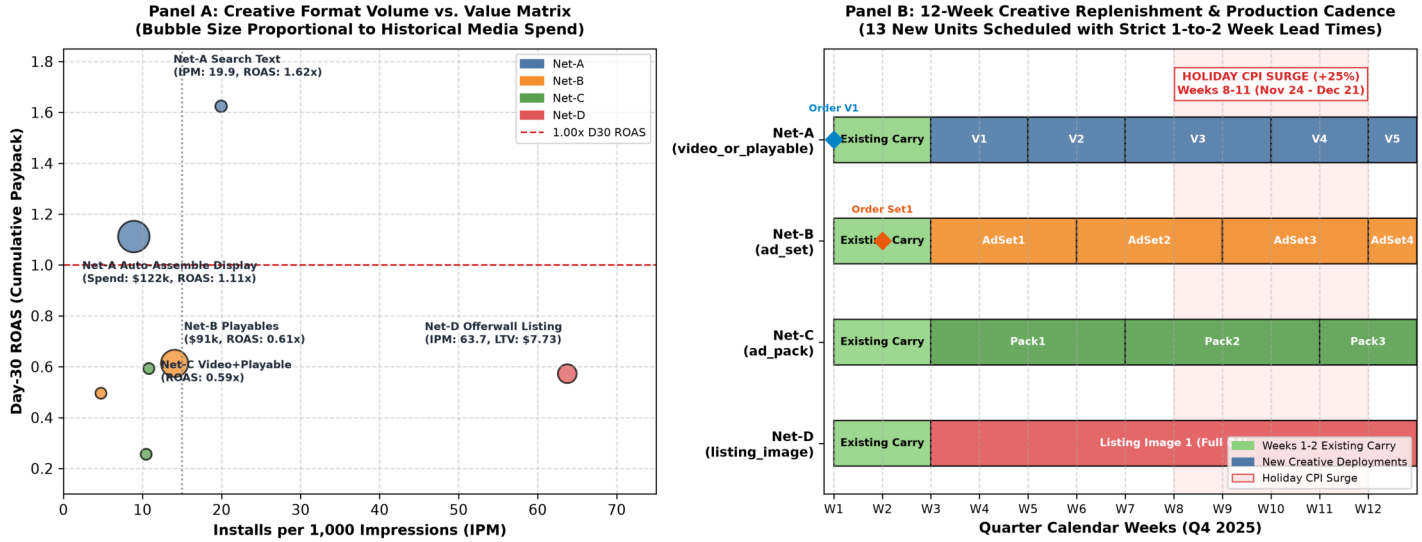


Exhibit 15: Creative Fatigue & Production Risk Simulation (Quantified Impact of Late/Insufficient Delivery)

Production Scenario	Quarter Spend	Units Built	Quarter Installs	Portfolio CPI ($)	D120 Gross Revenue	D120 Net Cash Loss	D120 ROAS	CFO Hurdle Status
Baseline (On-Time Replenis	$140,392	13	75,545	$1.86	$140,392	$0.00	100.00%	PASS
Scenario 1 (2-Week Product	$140,392	13	72,940	$1.92	$135,551	-$4,841.09	96.55%	FAIL
Scenario 2 (50% Under-Prod	$140,392	7	68,281	$2.06	$126,892	-$13,499.20	90.38%	FAIL

Executive Finding: A 2-week production delay erodes -$4,841 in cash and causes D120 ROAS to fail (96.55% < 100%). On-time creative delivery is a strict financial mandate.